

User Satisfaction Insight Report (v1)



Mobile Casual Games — Google Play public reviews
移动休闲游戏 · Google Play 公开评论 · BQ1-BQ5 一份完整报告

>
Source: 10,940 English reviews across 14 apps & 7 publishers (collected 2024-02 → 2026-05).
Generated from Day 1-5 pipeline outputs. See methodology in
docs/business_design/business_scope_three_layer_design.md.

Executive Summary / 摘要

EN — Six numbers a stakeholder can repeat tomorrow:

1. 14 apps, 7 publishers, 10,940 English reviews form the cross-section. Average adjusted rating is ~3.68 ★ (Bayesian, k = 200).
2. #1 Health Score: Penny & Flo (0.821); #14 Royal Kingdom (0.355) — a 2.3× spread within a single comparable category, showing the metric discriminates real satisfaction differences.
3. Royal Kingdom is the single largest portfolio risk: 60.7% of its reviews are 1-2★ and Dream Games' weighted publisher score (0.456) is the lowest of all 7 publishers.
4. Crash & Bug is the #1 fix-backlog driver across the entire pool: it pulls average rating down by +1.28★ wherever it appears, and 56.9% of those reviews are 1-2★.
5. 6 of 13 negative themes land in P0 of the BQ5 priority backlog. Critical-blocker themes (crash_bug, account_login, purchase_issue) are automatically promoted regardless of frequency.
6. Positive driver is real, not just noise: "Fun & Addictive" appears in 32.6% of all reviews, average 4.40★ — this is the moat to protect when shipping any change that touches difficulty or monetization.

CN — 六条带数字的汇报口径：

1. 14 个 App、7 家 Publisher、10940 条英文评论构成横截面；整体贝叶斯调整平均分约 3.68 ★。
2. 健康分第一：Penny & Flo (0.821) ；倒数第一：Royal Kingdom (0.355) —— 同品类内差距 2.3×，证明这个指标真的能区分体验差异。
3. Royal Kingdom 是组合里最大的风险源：60.7% 评论是 1-2★，所在的 Dream Games 加权 publisher 得分 0.456 也是 7 家最低。
4. Crash & Bug 是全池子第一号修复目标：一旦评论里提到，平均拉低 +1.28 ★，且 56.9% 是 1-2★。
5. 13 个负向主题中有 6 个进入 P0；阻断型主题（崩溃 / 登录 / 付款）即使频率不高也会自动升级 P0。
6. 正向驱动真实存在：Fun & Addictive 在 32.6% 评论里出现、平均 4.40★ —— 这是任何改难度或商业化前必须保护的护城河。

§1 Scope & Honesty / 数据范围与诚实声明

EN — All metrics in this report are review-based proxy signals of perceived satisfaction. The pipeline scrapes only English reviews from Google Play public pages: 10,940 rows after layered P0/P1/P2 cleaning. This is not a random sample, not the full user base, and not a substitute for retention / conversion / revenue / DAU.

CN — 本报告所有指标都是基于评论的感知满意度 proxy 信号。数据来源是 Google Play 公开页面、仅英文评论，分层 P0/P1/P2 清洗后保留 10940 条。这不是随机样本、不代表全部用户、不能替代留存 / 转化 / 收入 / DAU。所有结论需带这一句话作前提。

Item	Value
Apps in pool	14

Item	Value
Publishers	7
English reviews after cleaning	10,940
Date range	2024-02-12 → 2026-05-04
Global Bayesian mean (k = 200)	3.676

App Satisfaction Health Score — winners vs at-risk

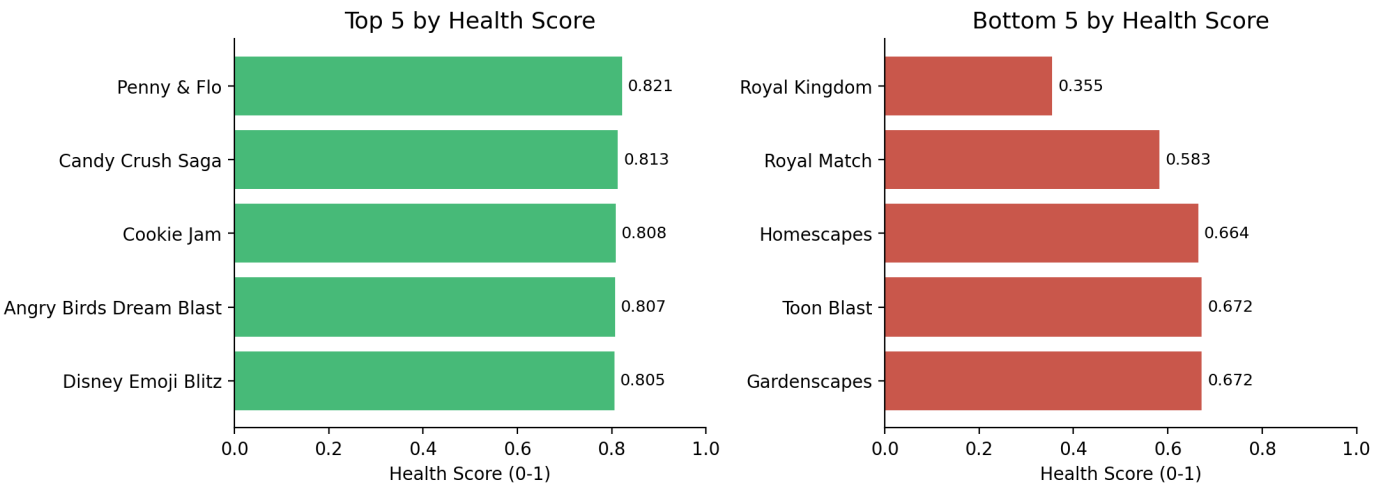


Figure: 01_top_bottom_health.png

§2 BQ1 — Overall Satisfaction Health / 整体满意度健康

EN — Health Score combines Level (40%, Bayesian-adjusted rating), Polarization (35%, low-star risk), Momentum (25%, recent rating trend), then multiplies by Confidence (review-volume based, capped at 1.0). Apps in the pool all carry confidence ≈ 1.0 (n ≥ 513).

- Top 3: Penny & Flo (0.821), Candy Crush Saga (0.813), Cookie Jam (0.808).
- Bottom 3: Homescapes (0.664), Royal Match (0.583), Royal Kingdom (0.355).
- Spread: 0.355 → 0.821 within a single comparable category — the metric is not flat.

CN — 健康分 = Level (40% , 贝叶斯调整后的均分) + Polarization (35% , 1 - 低星占比) + Momentum (25% , 最近趋势) , 最后乘 Confidence (按样本量上限 1.0) 。本池子所有 App 的 Confidence 都是 1.0 (每个 App 评论 ≥ 513 条) 。

- 前三 : Penny & Flo (0.821) 、 Candy Crush Saga (0.813) 、 Cookie Jam (0.808) 。
- 后三 : Homescapes (0.664) 、 Royal Match (0.583) 、 Royal Kingdom (0.355) 。
- 差距 : 0.355 → 0.821 , 同品类差距 2.3× , 说明指标是「真区分」而非平均化。

§3 BQ2 — Competitor / Publisher Benchmark / 竞品与公司层面对比

EN — Aggregating to publisher level (review-volume weighted) flips the storyline from "which app" to "which company's product culture is healthier":

- Best publisher: SGN/Jam City (0.808, 1 app — Cookie Jam).
- Worst publisher: Dream Games (0.456, 2 apps — Royal Match + Royal Kingdom).
- King and Rovio sit in the 0.79 band — multi-product portfolios with consistent execution.
- Volume-vs-Rating scatter identifies the diagonal where high review volume meets low rating — Royal Kingdom is the most isolated outlier.

CN — 用评论量加权聚合到 publisher 层面后 , 故事从"哪款 App"变成"哪家公司的产品体验更稳" :

- 最佳 : SGN/Jam City (0.808 , 1 款 — Cookie Jam) 。

- 最弱：Dream Games (0.456 , 2 款 — Royal Match + Royal Kingdom 都拖后腿)。
- King / Rovio 处于 0.79 一档 —— 多产品组合下执行依然稳定。
- 声量 vs 评分散点图里，Royal Kingdom 是「高声量 + 低评分」最孤立的离群点。

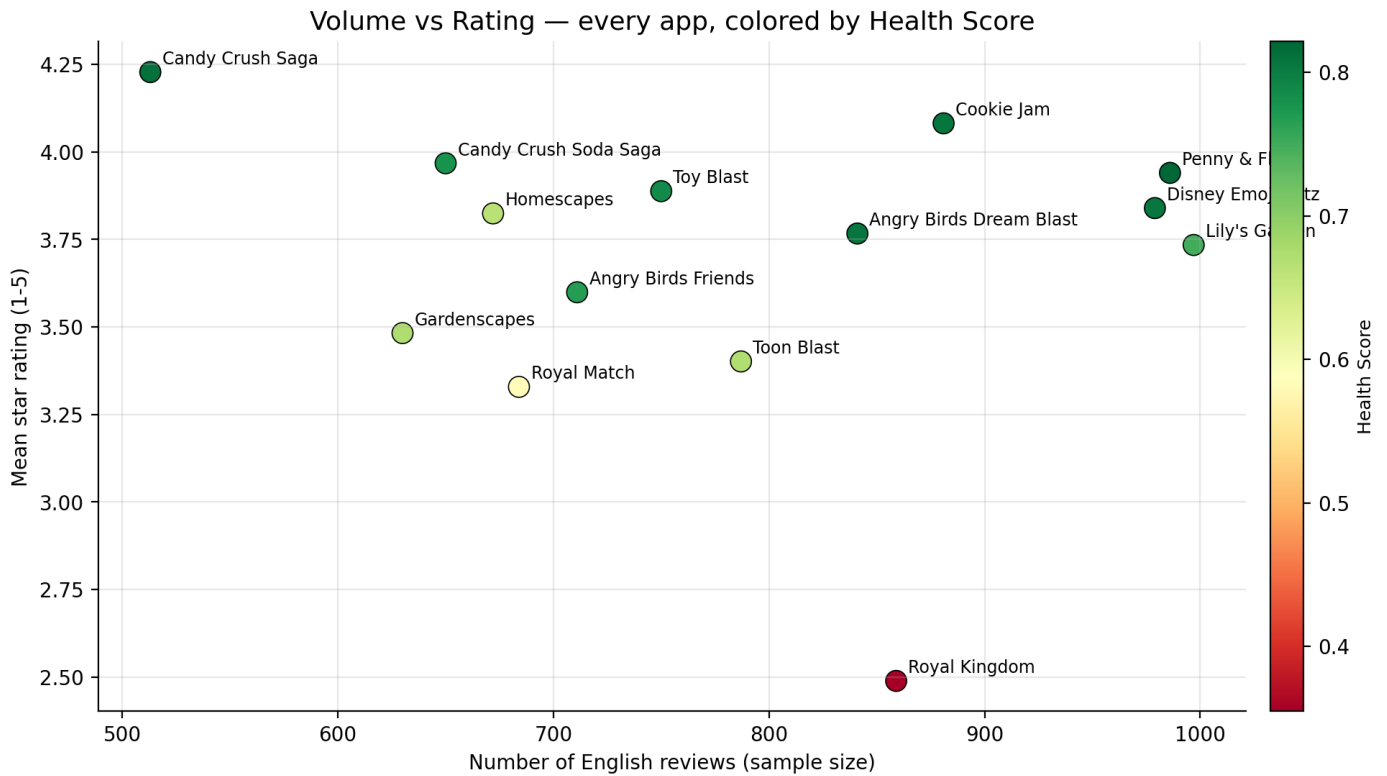


Figure: 02_volume_vs_rating_scatter.png

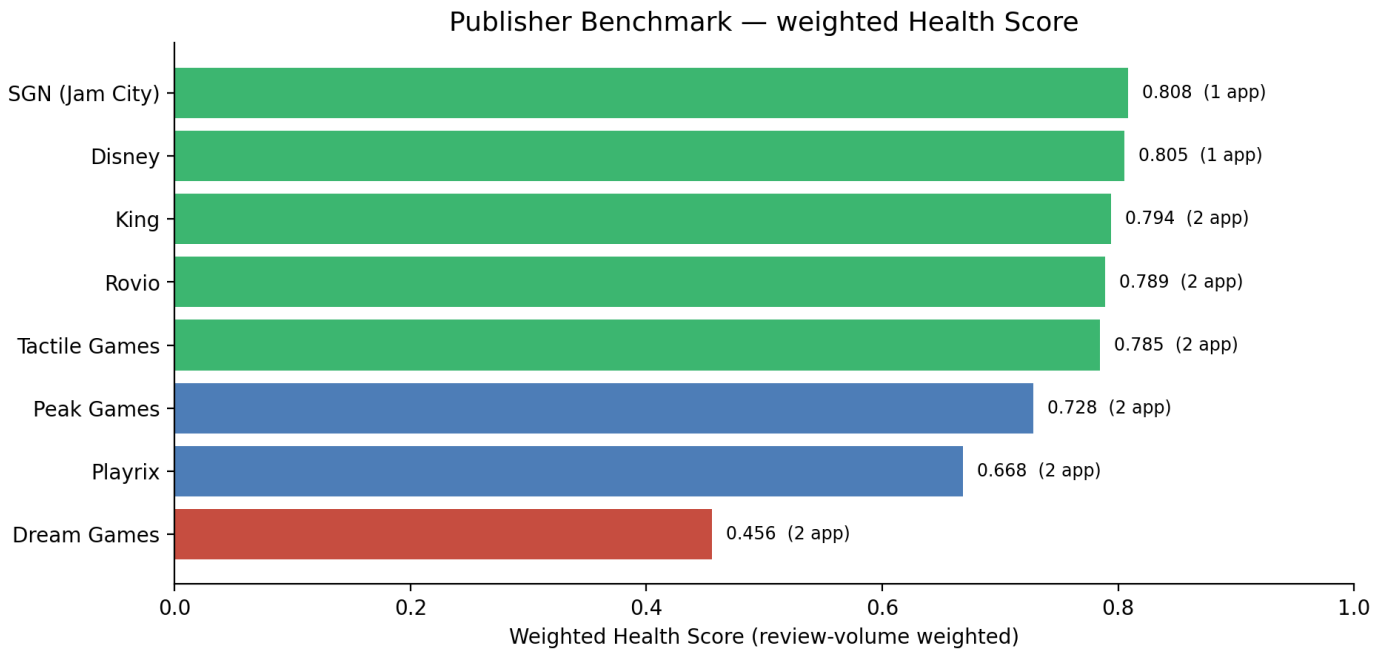


Figure: 03_publisher_weighted_health.png

§4 BQ3 — Pain Points / 低分痛点

EN — 18 themes (13 negative, 5 positive) were tagged via rule-based multi-label keyword matching; v2 dictionary cleared the obvious negation false positives (e.g. "used to be fun" no longer counts as positive). Theme coverage: 45.2% of reviews carry ≥ 1 theme.

Top rating-damaging negative themes (negative themes only):

Theme	n_reviews	rating_gap (★)	share_low_1_2
Crash & Bug	443	+1.28	56.9%
Loading & Performance	343	+1.08	51.9%
Difficulty & Progression	216	+1.03	52.3%
Pay-to-win / Greedy	154	+2.07	79.2%
Update Issue	90	+1.19	52.2%
Ads Volume	78	+1.43	61.5%

CN — 18 个主题 (13 个负向、5 个正向) 通过关键词规则多标签匹配；v2 字典已修掉典型的否定假阳性 (如 "used to be fun" 不再算正向)。主题覆盖率：45.2% 评论命中至少 1 个主题。

最"伤评分"的负向主题 (按 rating_gap 排)：上表。Crash & Bug 在全样本一旦被提，就拉低 1.28★，是头号工程修复对象；Pay-to-win 单次伤害最大 (2.07★)，且 79.2% 是 1-2★，需产品+经济侧统筹。

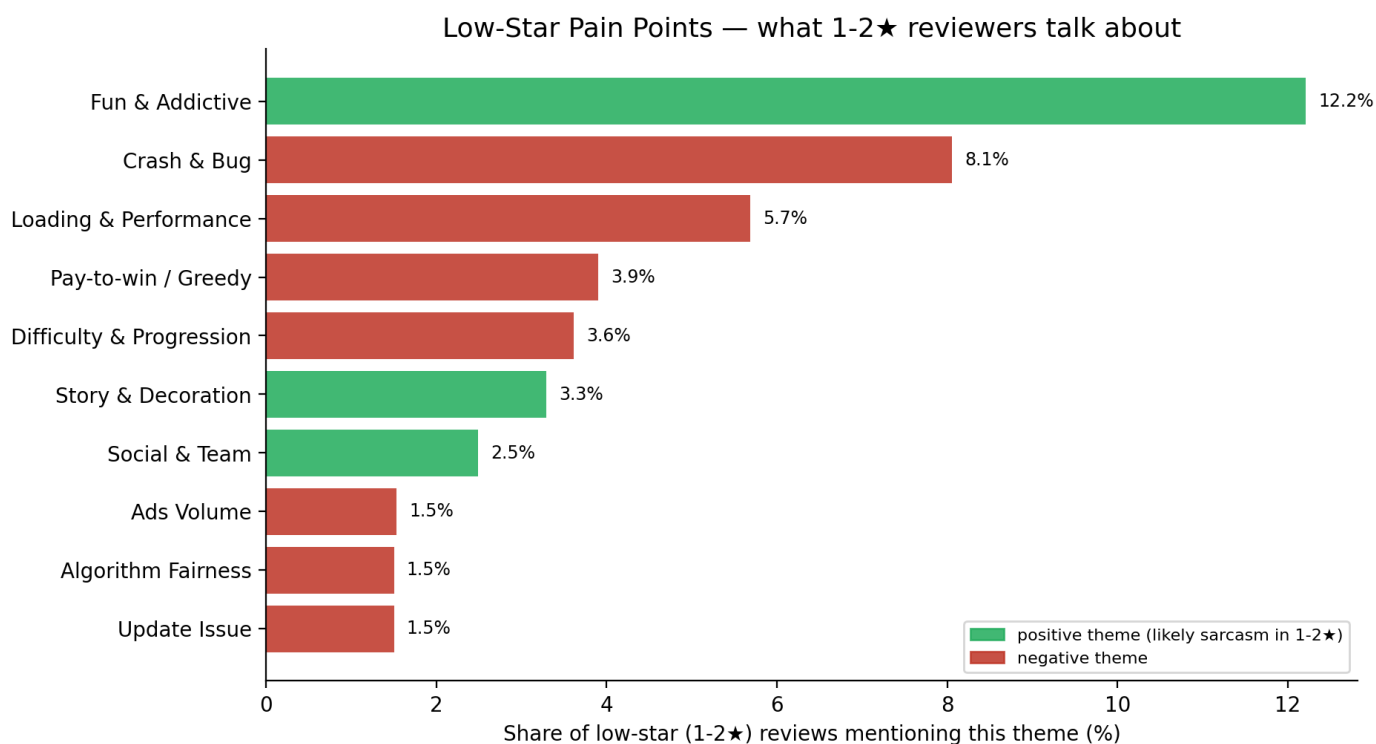


Figure: 04_low_star_theme_top10.png

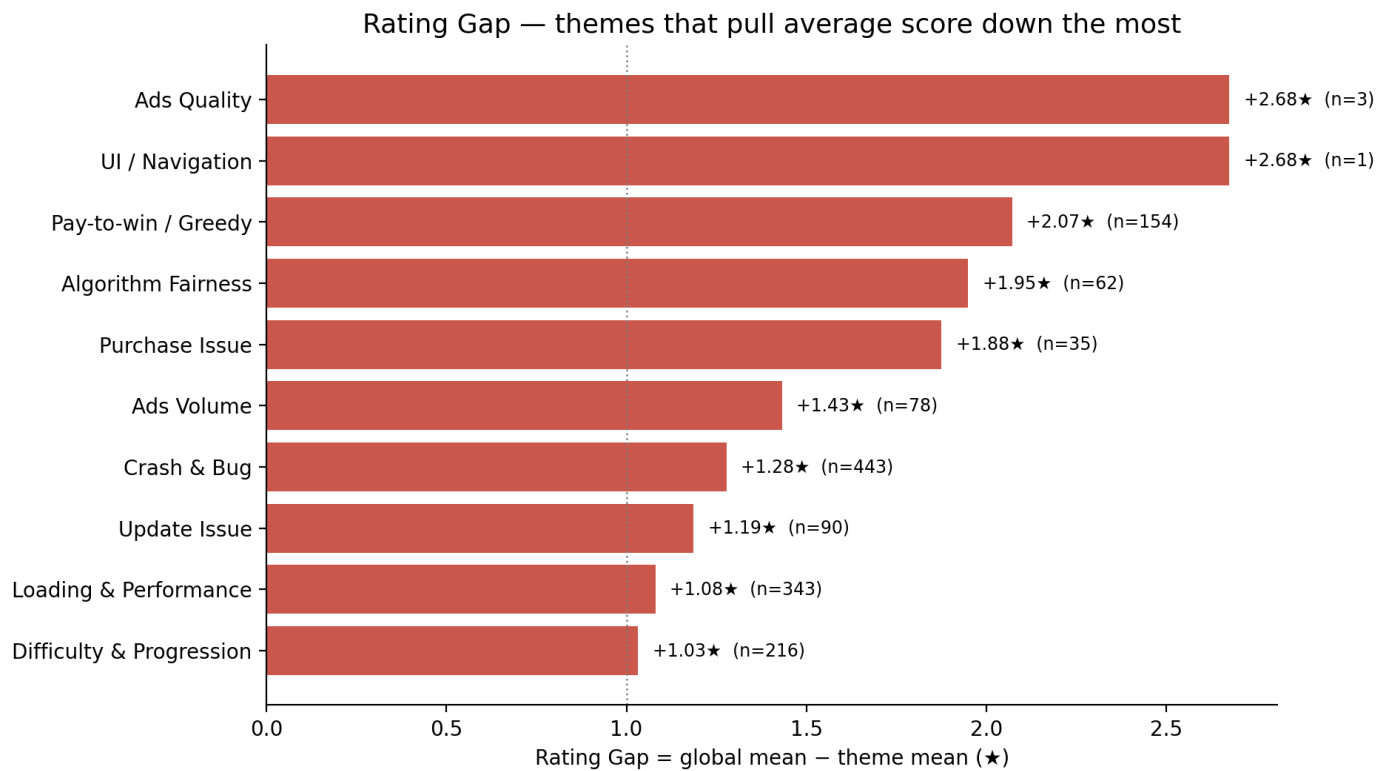


Figure: 05_rating_gap_top10.png

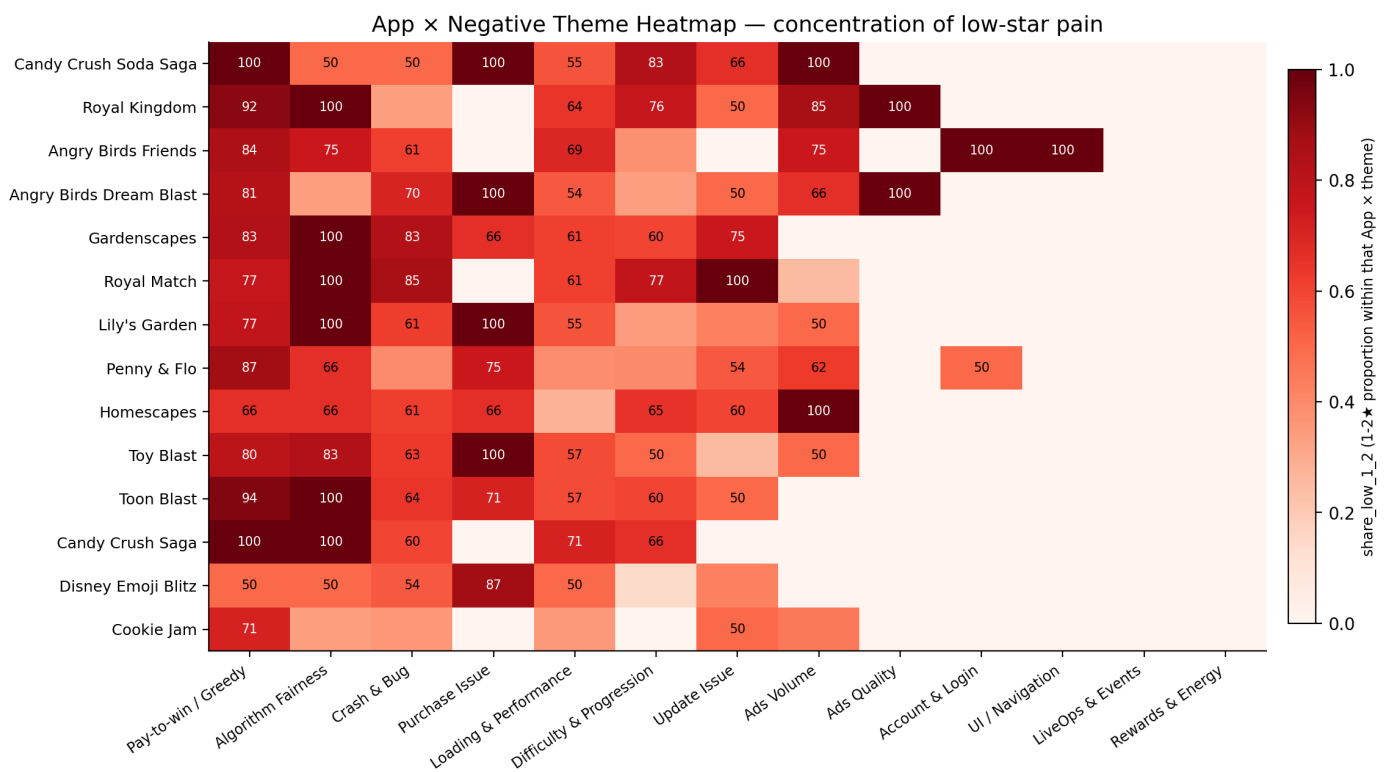


Figure: 06_app_theme_heatmap.png

§5 BQ4 — Positive Drivers / 高分驱动

EN — Don't treat the dashboard only as a complaint engine. The positive themes are the moat:

Theme	n_reviews	theme_share	mean_score
Fun & Addictive	3,567	32.6%	4.40 ★
Story & Decoration	519	4.7%	3.98 ★

Theme	n_reviews	theme_share	mean_score
Visual & Graphics	239	2.2%	4.57 ★
Social & Team	230	2.1%	3.43 ★
Updates & New Content	39	0.4%	3.46 ★

Royal Kingdom — what to protect even while fixing P0s: Fun & Addictive is still 18.7% of its in-app reviews; if monetization changes wipe that out, the recovery becomes harder.

Candy Crush Saga — case study of a stable positive moat: 30% of its reviews mention Fun & Addictive at 4.62★, with low-star share only 5.2%. The franchise endures because the core fun signal is so clean.

CN — 仪表盘不只是「投诉机器」。正向主题就是护城河：上表。

- Royal Kingdom —— 即便狠修 P0，也要保护：Fun & Addictive 仍在它内部评论里占 18.7%。
- Candy Crush Saga —— 稳定正向护城河的范本：30% 评论提到 Fun & Addictive、平均 4.62★，低星仅 5.2%。

§6 BQ5 — Priority Backlog / 优先级与产品建议

EN — Priority Score = normalized(Frequency) × normalized(Impact) × normalized(Severity) × 1.5^is_critical, where critical = {crash_bug, account_login, purchase_issue}. Quantile-based P0/P1/P2/P3 buckets are computed on negative themes only; positive themes are kept in the table but excluded from the fix backlog.

Negative theme distribution: P0 × 6, P1 × 2, P2 × 2, P3 × 3.

Bucket	Theme	Owner	Recommended Action	30-day goal
P0	Crash & Bug (critical)	Engineering	Crash triage + top-stack fixes	Reduce 1-2★ share by 10-15%
P0	Pay-to-win / Greedy	Product + Economy	Rebalance moves/booster/paywall pacing	Same
P0	Loading & Performance	Engineering	Cold-start & lag-spike optimization	Same
P0	Difficulty & Progression	Game Design	Tune curve, reduce hard-level clustering	Same
P0	Purchase Issue (critical)	Product + Economy	Fix payment failure / refund edge cases	Same
P0	Account & Login (critical)	Platform/Backend	Stabilize login / sync / progress	Same
P1	Algorithm Fairness	Game Design	Audit RNG perception, expose feedback	Scoped optimization, track rating gap
P1	Ads Volume	Product + Economy	Cut ad frequency in early progression / fail loops	Scoped optimization, track rating gap

CN — 优先级公式上面已经说明。负向主题分布：P0×6, P1×2, P2×2, P3×3。表中给出每个 P0/P1 项的 owner、建议动作、30 天目标，可直接进 backlog。

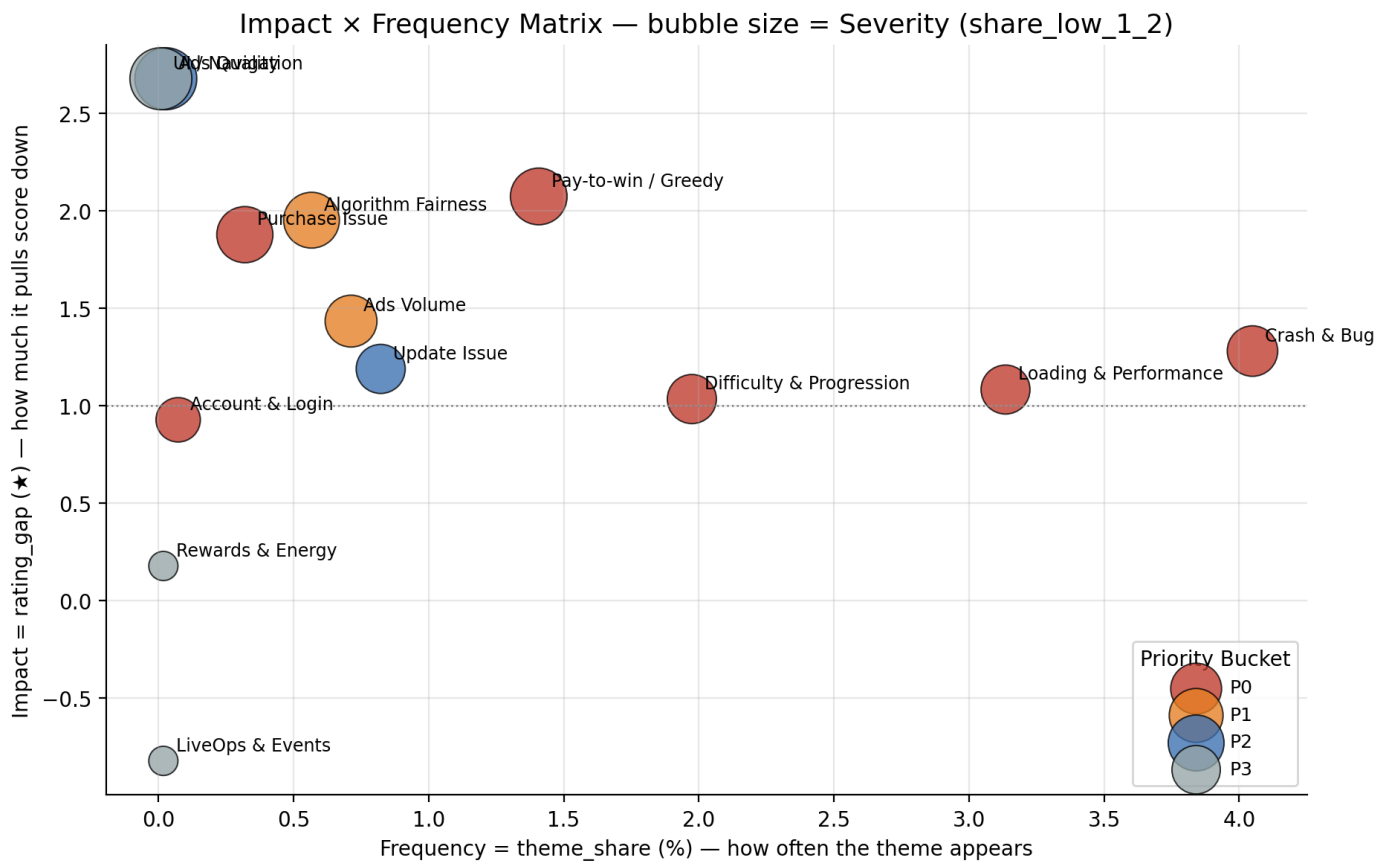


Figure: 07_impact_frequency_matrix.png

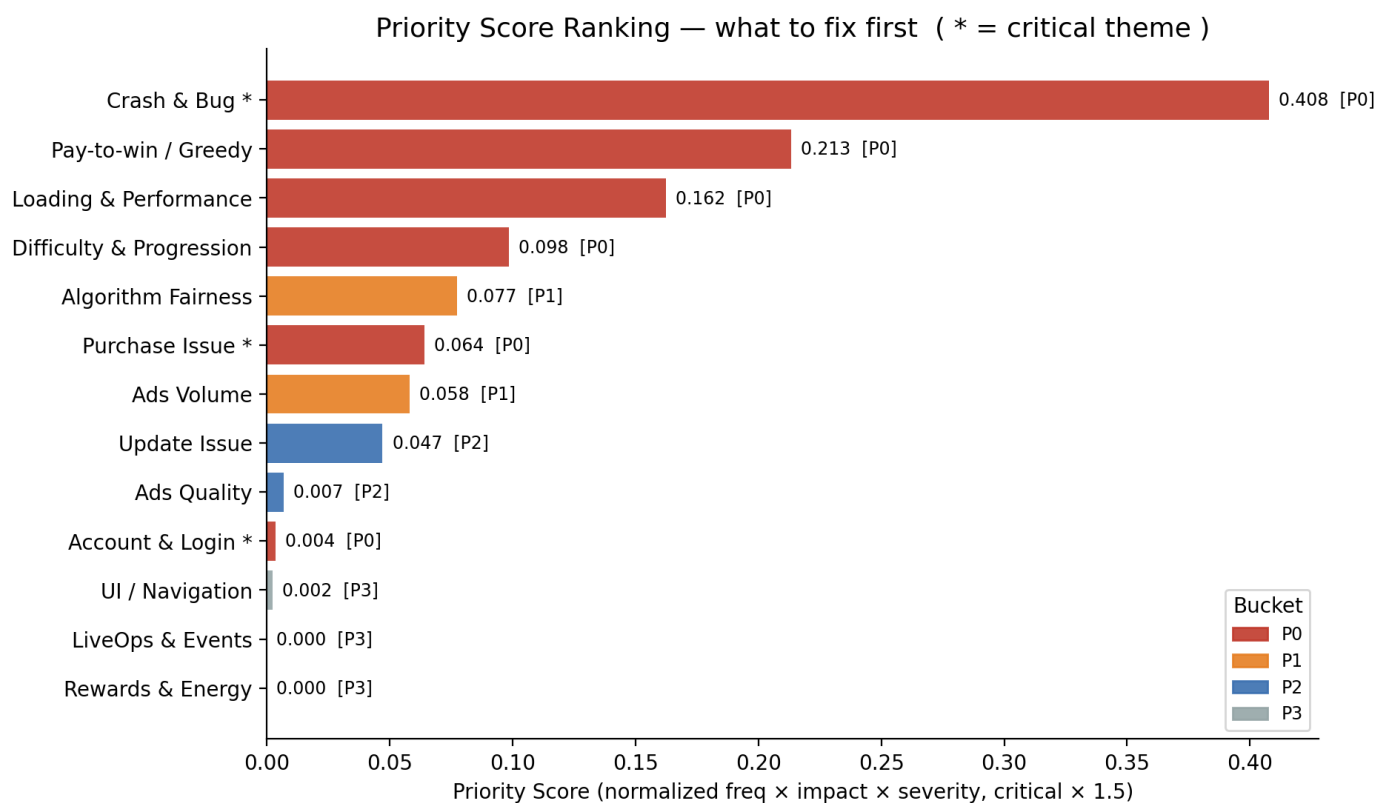


Figure: 08_priority_score_ranking.png

§7 Limitations & Data Honesty / 局限与诚实声明

EN —

1. Sampling: Google Play public reviews ≠ random sample of users. Sort order, recency, and language filters all bias the pool.
2. English-only: ~65% of cleaned-stage reviews. Non-English subgroup excluded — markets like JP/KR may differ.
3. Proxy not KPI: Health Score and Priority Score reflect *perceived* satisfaction. They are not retention, ARPU, conversion, or DAU.
4. Rule-based tagging: 18-theme keyword dictionary has known limitations on sarcasm, mixed sentiment, and unique phrasings. Coverage of 45.2% means a meaningful portion of reviews remains unclassified — those should not be assumed to be empty of signal.
5. Theme overlap: Multi-label tagging means one review can drive multiple theme metrics — counts are not mutually exclusive.
6. Cross-section, not causal: This report describes the current state. It does NOT claim any of the recommendations will cause specific rating uplift; the 30-day goals are working targets, not predictions.
7. No version / event-window analysis in v1. Appendix G in the design doc reserves this for Phase 2.

CN —

1. 抽样：Google Play 公开评论 ≠ 随机样本，受排序、时效、语种过滤影响。
2. 仅英文：清洗后约 65% 评论。日韩等市场未覆盖。
3. 是 Proxy 不是 KPI：Health / Priority 反映「感知满意度」，不是留存 / ARPU / 转化 / DAU。
4. 规则字典局限：18 个主题对反讽、混合情绪、独特表达识别能力有限；45.2% 覆盖率意味着仍有大量评论未被分类，不能视为“无信号”。
5. 多标签可重复计：一条评论会贡献到多个主题指标，计数不互斥。
6. 横截面 + 描述性：本报告描述现状，不承诺任何建议必然带来评分提升。30 天目标是工作目标，不是预测。
7. 未做版本 / 事件窗口分析：留作 Phase 2 (设计文档 Appendix G) 。

§8 Next Steps (Phase 2) / 下一步

EN —

- LLM-assisted theme tagging for the unclassified 54.8% (and for sarcasm/mixed-sentiment edge cases).
- iOS App Store expansion for cross-platform satisfaction comparison.
- Internal metric integration (retention, IAP, ad engagement) to validate review-based proxies.
- Event / version window analysis once reviewCreatedVersion coverage ≥ 50%.
- Causal evaluation (A/B, regression discontinuity) once a fix actually ships and we can read pre/post evidence.

CN — 见 EN 同条；按设计文档 Future Work 路线推进。

§9 Methodology Pointers / 方法论速查

Topic	File
Cleaning layers (P0/P1/P2)	scripts/02_clean/clean_and_eda.py
Health Score formula	scripts/06_insights/build_health_score.py + config/metrics.json + design doc Appendix A/B/C/D
Theme taxonomy	config/themes.yml (v2) + design doc §11
Theme tagging engine	scripts/06_insights/build_review_themes.py
Theme summaries	scripts/06_insights/build_theme_summary_tables.py
Priority Score	scripts/06_insights/build_priority_recommendation_tables.py + design doc §10.5
Daily logs	docs/daily_logs/day{1..6}_*.md
Plain-language briefings	reports/■■■■/Day{4,5}_*.pdf
This report (regenerate)	python3 scripts/04_export/build_insight_report_pdf.py

